Vignesh G

Senior Software Developer | Chennai, India
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Summary

Over 6+ years of professional backend development experience using Python, with deep expertise in Django, Django Rest Framework, and FastAPI for building scalable REST APIs and microservice architectures.

Skilled in designing event-driven, high-concurrency systems using Apache Kafka, Redis, and API gateways (APISIX, KrakenD), supporting platforms with 50,000+ users and 100,000+ concurrent connections.

Experienced with relational and vector databases (PostgreSQL, MySQL, SQLite, Qdrant), multi-tenant schema design using Django-Tenants, and cloud-native authentication with AWS Cognito and RBAC.

Hands-on experience building AI-powered applications using LangChain, OpenAI, and Agentic RAG pipelines, alongside containerized deployment with Docker and CI/CD optimization.

Strong knowledge of the Django MVT pattern, ORM design, version control (Git/GitHub), and the ODOO Python framework for customized ERP/HRMS modules.

Experience

Yitro Technologies Pvt Ltd Chennai, India

Senior Backend Developer *Oct 2023 – Present*

- **SmartAlly Tutor (AI-Powered E-Learning Platform):** Designed scalable FastAPI microservices for course management, enrollment, assessments, certification, and AI-powered learning workflows.
- Built an Agentic Retrieval-Augmented Generation (RAG) system using LangChain, OpenAI, and Qdrant for context-aware, course-specific tutoring, with semantic search over vector embeddings for accurate content retrieval.
- Developed real-time conversational AI with streaming responses and voice-based chat, and built AI agents to automate course content generation, question paper creation, and assessment preparation.
- Designed document ingestion pipelines (chunking, embedding generation, vector indexing) and integrated Redis caching/session management with Kafka-based asynchronous event processing for scalability; containerized services with Docker for consistent deployment.
- **Pass (Super App Platform):** Contributed to a scalable, microservice-based Super App integrating ride booking, package delivery, and e-commerce; optimized backend services with Python, FastAPI, and PostgreSQL, reducing API latency by 25% across order, delivery, and user management systems.
- Designed isolated gRPC-based microservices glued together by FastAPI services, enabling efficient inter-service communication across the platform.
- Used Redis to power live location tracking and nearby driver discovery, and implemented WebSocket-based real-time chat between riders and drivers.
- Implemented Kafka-based event-driven communication and Redis caching to improve scalability and asynchronous processing; configured APISIX and KrakenD API gateways for secure routing, authentication, and traffic management across modules.
- Developed secure authentication/authorization workflows using AWS Cognito with RBAC, and enabled secure payment processing through Stripe, Razorpay, and GTI Payments across ride, delivery, and shopping services.
- **Jobconnectz:** Developed high-performance microservices using Python and FastAPI supporting user management, payments, and booking workflows for over 50,000 users; architected independently scalable services, enabling 30% faster feature releases.
- Containerized all services with Docker and optimized CI/CD pipelines, shortening deployment time by 40%; configured APISIX API Gateway for routing, load balancing, and rate limiting, reducing API response times by 25%.
- Strengthened authentication using AWS Cognito with OAuth and social login (increasing login success rates by 15%) and established RBAC policies securing sensitive data across 3 user groups.
- Leveraged Firebase Cloud Messaging (FCM) for real-time notifications, boosting user and provider engagement by 20%; optimized PostgreSQL schemas, cutting query latency by 35% for key workflows.
- Spearheaded an asynchronous communication architecture with Apache Kafka, enhancing platform stability and enabling seamless scaling to 100,000 concurrent users.

Ciranta IT Services Pvt Ltd Chennai, India

Senior Software Developer *Oct 2020 – Sep 2023*

- **ERP Software:** Built a microservice-based ERP system for a leading Indian jewellery retail chain, covering Sales, Purchase, Inventory, and Reporting modules, using Angular, Django Rest Framework, and MySQL.
- Gathered requirements directly from clients, designed and tested REST APIs before production deployment, documented API requests for other developers, implemented the Inventory module, and built automated reports using Pandas and XlsxWriter.
- **Customer Payment Portal (Kalyan Portal):** Developed the backend for an enrolled gold scheme handling advance payment and monthly payments, using React, Django Rest Framework, and PostgreSQL.
- Implemented user authentication for SignUp and Login using DRF, integrated a payment gateway, and worked on React forms, list views, and API integration.
- **Africa Institute (HRMS in ODOO):** Built an HRMS system in the ODOO framework that maintains employee information, records daily attendance, and manages annual appraisals, using Odoo and PostgreSQL.
- Developed customized modules for travel request and annual feedback forms, asset creation/purchase/transfer, and helpdesk; supported clients via email and live product demos.
- Collaborated closely with the frontend Angular team on end-to-end project and database architecture design, and tracked bugs/documented issue resolutions using Atlassian Jira and Confluence.

Flendzz Technologies (P) Ltd Chennai, India

Associate Application Developer *Jun 2019 – Mar 2020*

- **Gibbous (SaaS Payroll Software):** Developed a SaaS-based HR payroll platform enabling payroll processing, employee management, ESS portal, and attendance tracking with automatic updates and improved transparency, using Angular and Django Rest Framework.
- Implemented payroll processing functionality with updated statutory compliances and tax regulations, and built/integrated new and existing API endpoints using Django Rest Framework.
- Collaborated with the frontend Angular team on the overall project and database architecture, and performed bug tracking and issue resolution documentation using Atlassian Jira and Confluence.

Samrantech Pvt Ltd Chennai, India

Software Developer Trainee *Dec 2018 – May 2019*

- Gained hands-on foundational experience in Python and Django web application development.

Education

SRM Valliammai Engineering College Chennai, India

Bachelor of Engineering in Electronics & Communication Engineering

Technical Skills

Languages: Python

Frameworks: Django, Django Rest Framework, FastAPI, ODOO

Databases & Storage: PostgreSQL, MySQL, SQLite, Redis, Qdrant

Messaging & Gateways: Apache Kafka, APISIX, KrakenD, gRPC, WebSocket

Cloud, Auth & DevOps: AWS Cognito, OAuth, RBAC, Docker, CI/CD, Firebase Cloud Messaging (FCM)

AI/ML: LangChain, OpenAI, Agentic RAG, Vector Embeddings

Payments: Stripe, Razorpay, GTI Payments

Libraries & Tools: Pandas, XlsxWriter, Wkhtmltopdf, Git/GitHub, HTML & CSS

Operating Systems: Windows, Ubuntu, Linux Mint

Languages Spoken: Tamil, English

Recognitions

Synergic Force Award – Yitro Technologies Pvt Ltd (2024, 2025)

Employee of the Month – Ciranta IT Services Pvt Ltd (November 2021, December 2021)